

TASK 2

ICT703: Programming

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Figure 1: Where to caption a figure..... **Error! Bookmark not defined.**

1 Introduction

Briefly introduce the purpose of the program and the problem being solved.

2 Problem Description

- o The purpose of your program
- o The main functionality provided
- o The transaction data used
- o Any assumptions made
- o Any known limitations

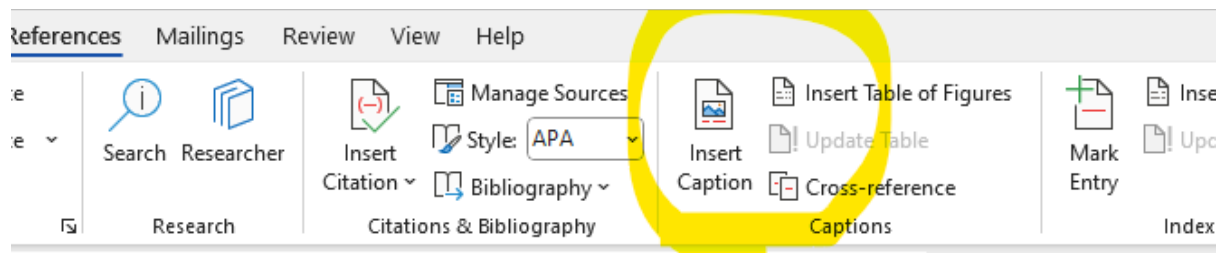
3 Program Design

Explain the overall design of your solution. Include:

- o The overall program structure
- o Major functions and their purpose
- o Data structures used (for example, lists or dictionaries)
- o Pseudocode or flowcharts illustrating your solution

You may include your initial design, final design, or both

Figure 1: Where to caption a figure.



4 Libraries and Dependencies

Identify any Python libraries used in your program.

For each library, include:

- o Library name
- o Purpose
- o Installation instructions (if required).

If no additional libraries are required beyond the Python Standard Library, state this clearly.

5 Fraud Detection Rules

Explain how the three core fraud-detection rules were implemented.

If advanced features were completed, briefly explain how they extend the core functionality.

6 Advanced Features

If you attempted any advanced features, concisely explain:

- o Which advanced feature(s) you selected
- o How each feature was implemented; any limitations or incomplete elements
- o How each feature extends the core program
- o Any additional data files, libraries, configuration files, or database components used

Include screenshots, diagrams, short code examples, or sample outputs where appropriate. If no advanced features were attempted, state this clearly.

7 Testing

Include:

- o A test plan
- o Detailed test cases
- o Test inputs
- o Expected outputs
- o Actual outputs
- o Test results (Pass/Fail)

Screenshots of selected program output may also be included where appropriate.

8 Challenges and Future Improvements

Briefly discuss:

- o Challenges encountered during development
- o How the challenges were addressed
- o Possible future improvements to the program

9 Reflection on Artificial Intelligence (AI) Use

Briefly explain how AI tools were used during the planning, development, debugging, testing, documentation, or report-writing stages.

Include:

- o The AI tool(s) used
- o At least three detailed prompts and a brief summary of the corresponding outputs
- o How each output was applied, modified, or rejected
- o Any errors, limitations, or unsuitable suggestions identified
- o How AI-generated content was checked, tested, and verified

Where relevant, screenshots of AI prompts and outputs can be included to illustrate your reflection.